The UK Linguistics Olympiad 2017

Round 2 Problem 5. Magic Yup'ik

Central Alaskan Yup'ik belongs to the Eskimo-Aleut language family. It is spoken in western and southwestern Alaska by around 20,000 speakers. Yup'ik people have an interesting approach to counting: the words for the numbers can be broken down into meaningful parts which may be related to their body parts. For example, the word for five, *talliman*, means *one arm* and the word for six, *arvinlegen*, means *cross over*, as you need to change hand to go on counting.

Another interesting fact about the Yup'ik people is that their parkas often include geometrical patterns such as the 3x3 square shown below.

	1	2	3
a			
b			
c			

This problem is about an imaginary 'magic square' based on this pattern. In a magic square with nine cells, like this one, every cell contains one of the digits 1 to 9, every one of these digits is found in just one cell, and the sum of all the digits in every row, column, and diagonal is the same. In our magic square, the middle cell of the top row (i.e. cell 2a) contains the digit 9.

The clues in the next table will help you fill in the rest of the cells by defining each row and each column as a three-digit number which could also be defined in words as a complex number-phrase. For example the sequence of digits 1-2-3 defines the number 123, which can also be given in English by the number-phrase "One hundred and twenty three". The only snag is that the number-phrases in the table are in Yup'ik. HINT: The Yup'ik name for the number 294 is "yuinaat qula cetaman qula cetaman".

Across		Down	
a	Yuinaat yuinaq cetaman qula malruk	1	Yuinaat yuinaq atauciq akimiaq pingayun
b	Yuinaat akimiaq malruk akimiaq malruk	2	Yuinaat yuinaak malruk yuinaat malrunglegen qula atauciq
c	Yuinaat yuinaak malruk akimiaq atauciq	3	Yuinaat qula pingayun akimiaq atauciq

Q.5.1. Fill in the magic square on your answer sheet.

Q.5.2. Give the Yup'ik number-phrase for the number defined by the diagonal of cells 1a-2b-3c.

Q.5.3. Explain your answers.

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Problem 2.5 Magic Yup'ik (answer blank)

Q.5.1

	1	2	3
a		9	
b			
c			

Q.5.2 1a + 2b + 3c =

Q.5.3 Explanation (if you need more paper, ask for it – don't write on the back of this sheet!)